

Executive Summary for Gamification Team

We resolved task collision conflicts during artisan approval and stabilized the quest provisioning pipeline. **Slots 1 and 2 are now strictly reserved as canonical system baseline tasks** for every artisan workshop quest. All artisan custom activities automatically start at **Slot 3 or higher** without exception.

1. CANONICAL QUEST TASK SLOT STANDARD

To provide a consistent gamified journey for tourists visiting cultural workshops across Melaka and Malaysia, every approved artisan quest adheres to the following slot hierarchy:

Slot	Task Title	Type	XP Reward	Tourist Gamification Mechanism
Slot 1	Go to the workshop	MANDATORY SYSTEM	50 XP	GPS Geofence check-in (unlocked within 50m radius of artisan studio coordinates).
Slot 2	Stay for 15 minutes	MANDATORY SYSTEM	50 XP	Dwell timer tracking (900s) + Master QR scan verification (WK_ARTISAN:id:secret).
Slot 3+	Custom Artisan Activities	ARTISAN CUSTOM	50 – 200 XP	Optional activities created by the master artisan (e.g., "Try pottery wheel", "Craft photo").

2. WHAT CHANGED IN THE GAMIFICATION VIEWMODEL

File: lib/viewmodels/gamification_viewmodel.dart (Method: addHeritageTask)

Previously, when an artisan added a custom activity, the next sort order was calculated purely as highestSortOrder + 1 . If no tasks existed yet, it assigned sort_order = 1 , directly colliding with mandatory system tasks.

The Fix: Slots 1 & 2 are permanently reserved for system tasks:

```
// Reserved slot allocation: Custom tasks always begin at Slot 3 or above
final newSortOrder = highestSortOrder < 2 ? 3 : highestSortOrder + 1;

await _repository.addHeritageTask(
  questId: quest.id,
  title: cleanTitle,
  isRequired: isRequired,
  xpReward: xpReward,
  sortOrder: newSortOrder, // Guaranteed ≥ 3
);
```

3. AUTO-SHIFT & NORMALIZATION IN SUPABASE SERVICE

File: lib/data/services/supabase_service.dart (Method: _ensureDefaultSystemTasksForApproval)

1. Auto-Shift for Conflicting Custom Tasks
2. Baseline In-Place Normalization

If legacy custom tasks already occupy Slot 1 or 2, the approval engine automatically moves them to Slot 3+ (e.g., sort_order 3, 4, 5). **No artisan tasks are deleted or corrupted.**

Recognizes trigger-created baseline tasks (matching "workshop" and "15 min") and normalizes `is_system_task: true` in-place, eliminating redundant `INSERT` calls.

4. DATABASE RLS POLICIES REQUIRED FOR GAMIFICATION TABLES

The `heritage_tasks` and `quests` tables have Row-Level Security enabled. A migration script (`migration_fix_heritage_tasks_rls.sql`) has been prepared for the Supabase SQL editor to ensure administrators and tourists have proper permissions:

- `heritage_tasks` : Granted `SELECT, INSERT, UPDATE, DELETE` to `anon, authenticated, service_role` .
- `quests` : Granted `SELECT, INSERT, UPDATE, DELETE` to `anon, authenticated, service_role` .
- **RPC Integration:** `admin_update_user_status` runs as `SECURITY DEFINER` to atomically approve quests and normalize default journey tasks upon artisan approval.

5. HOW THIS AFFECTS YOUR GAMIFICATION WORKFLOW

- **Quest Progress & Stamps:** No breaking changes to `quest_progress` , `task_progress` , or `passport_stamps` . Tourists still receive stamps upon completing all tasks in a quest.
- **UI Display Order:** Tasks will consistently display in order: **(1) Arrival → (2) 15–Minute Dwell → (3+) Custom Workshops**.
- **Admin Moderation:** Admin approval will never fail due to task order collisions or RLS permission errors.

Action Required for Gamification Teammates

Please pull the latest commit from branch `origin/Tze-Chuen` (commit `3b33539`). When testing custom task creation in the Artisan Studio tab, note that all new custom tasks will display sequence numbers starting at **3**.